

## 5.5.2 Electromagnetic radiation from stars

| Learning outcomes                                                                                                                     | Additional guidance                            |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                         |                                                |
| (a) energy levels of electrons in isolated gas atoms                                                                                  | <b>THIS WILL ALSO REQUIRE KNOWLEDGE OF 4.5</b> |
| (b) the idea that energy levels have negative values                                                                                  |                                                |
| (c) emission spectral lines from hot gases in terms of emission of photons and transition of electrons between discrete energy levels |                                                |
| (d) the equations $hf = \Delta E$ and $\frac{hc}{\lambda} = \Delta E$                                                                 |                                                |
| (e) different atoms have different spectral lines which can be used to identify elements within stars                                 |                                                |

- (f) continuous spectrum, emission line spectrum and absorption line spectrum
- (g) transmission diffraction grating used to determine the wavelength of light
- (h) the condition for maxima  $d\sin\theta = n\lambda$ , where  $d$  is the grating spacing
- (i) use of Wien's displacement law  $\lambda_{\max} \propto \frac{1}{T}$  to estimate the peak surface temperature (of a star)
- (j) luminosity  $L$  of a star;  
Stefan's law  $L = 4\pi r^2 \sigma T^4$   
where  $\sigma$  is the Stefan constant
- (k) use of Wien's displacement law and Stefan's law to estimate the radius of a star.

The structure and use of an optical spectrometer are not required;

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Proof of this equation is not required.

**THIS WILL ALSO REQUIRE  
KNOWLEDGE OF 4.4.1**

